Research article

De novo **profiling of long non-coding RNAs involved in MC-LR–induced liver injury in whitefish: discovery and perspectives**

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Abstract

Background: Microcystin-LR (MC-LR) is a potent hepatotoxin for which a substantial gap in knowledge persists regarding the underlying molecular mechanisms of liver toxicity and injury. Although long non-coding RNAs (lncRNAs) have been extensively studied in model organisms, and their roles have been identified in various cellular processes including participation in regulation of gene expression together with microRNAs, our knowledge concerning the role of lncRNAs in liver injury is limited even in mammals. Given that lncRNAs show low levels of sequence conservation, their role becomes even more unclear in non-model organisms without an annotated genome, like whitefish *(Coregonus lavaretus)*. The objective of this study was to discover and profile aberrantly expressed polyadenylated lncRNAs that are involved in MC-LR–induced liver injury in whitefish.

Results: Using polyA-enriched RNA-Seq data, we *de novo* assembled a high-quality whitefish liver transcriptome. This enabled us to find 94 differentially-expressed (DE) putative evolutionary-conserved lncRNAs (orthologous to known lncRNAs in other species), such as MALAT1, HOTTIP, HOTAIR or HULC and 4,429 DE putative novel whitefish lncRNAs, which differed from annotated protein-coding transcripts (PCTs) in terms of corrected minimum free energy, GC base-pair content and length. Additionally, we identified DE non-coding transcripts that might be 3' autonomous untranslated regions of mRNAs (3'UTRs). We found that, in response to MC-LR treatment, these potential 3'UTRs could either be coexpressed with PCTs from the same mRNA, or DE in opposite directions, suggesting different 3'UTR-dependent gene regulation.

Z komentarzem [AR1]: "length-corrected" or maybe just delete the word?

— **sformatowano:** Nie Wyróżnienie

— **sformatowano:** Wyróżnienie

Z komentarzem [AR2]: If you are referring to the hypothesis of Malka et al. about global gene regulation by 3'UTR shortening, rewrite as "…, or the 3'UTRs were upregulated while the corresponding PCTs were downregulated, suggesting 3'UTR-dependent gene regulation."

If you weren't referring to that hypothesis, I don't understand what you're trying to say here.

Conclusions: To our knowledge, this is the first report on aberrantly expressed lncRNAs in MC-

LR–induced liver injury in whitefish. We found both evolutionary conserved lncRNAs as well as novel whitefish lncRNAs that could serve as biomarkers of severe and chronic liver injury. The lncRNA sequence data files and the raw sequence files are available in the web supplement and the NCBI Sequence Read Archive, respectively.

Background

A substantial gap in knowledge persists regarding the role of microcystins (MCs) in the underlying molecular mechanisms of organ toxicity and injury. MCs are a group of cyclic heptapeptide hepatotoxins, of which microcystin-LR (MC-LR) is one of the most widely distributed and potent variants. MC-LR is absorbed, transported and accumulated predominantly in the liver [1], and it causes drug-induced liver injury (DILI). Studies on the transcriptomic level have revealed various protein coding transcripts (PCTs) involved in the response and progression of MC-LR-induced liver injury in different species [2]. In addition to PCTs, various non-coding RNA transcripts (ncRNAs, NCTs) have been implicated in the responses to various stressors, including [4]. MC-LR alters the expression levels of small regulatory ncRNAs (shorter than 200 nt) like microRNAs (miRNAs), piwi-associated RNAs (piRNAs) and small interfering RNAs (siRNAs) [5] in various types of tissues and cells [6, 7].

In comparison to knowledge about small regulatory ncRNAs, our understanding of the functions and mechanism of action of long non-coding RNAs (longer than 200 nt; lncRNAs) is still limited. Unlike miRNAs, lncRNAs are poorly conserved among species [8], which hinders research on their function and evolution. However, it has been shown that lncRNAs are involved in a variety of biological processes such as cell proliferation, apoptosis and differentiation [9] by regulating gene expression via a variety of mechanisms including binding (sponging) miRNAs. In sponging, lncRNA competitively binds to miRNA, resulting in changes in the protein level of coding genes at the post-transcriptional level [10]. lncRNAs may function as competing endogenous RNAs (ceRNA) that share common miRNA-response elements with PCTs [11]. For example, the recently characterized metastasis-associated in lung-adenocarcinoma transcript-1

Z komentarzem [AR5]: Do you really need both acronyms? It seems to just add unnecessary confusion for the reader, unless I'm missing something. Generally speaking, it is better to use one abbreviation in a consistent way whenever possible.

Abbreviations should be used consistently whenever possible. Hofmann (2014) writes: "Once you have defined an abbreviation, use it whenever you need it—do not switch back to using the full term unless many pages have elapsed since its previous appearance. Then you may remind the reader, once, what the abbreviation means." Zeiger (2000), Schultz (2009), and many others give similar advice. Good English style for an international audience is different from Polish style. So, even if you are uncomfortable repeating a key word or an abbreviation, just do it—it's good international English style and easier to understand, and it makes reviewers happy.

I've improved your usage of abbreviations whenever I noticed a problem, but I have not done a Find-Replace throughout the text—I figure you can do that yourself and you don't need to pay me to do it. So, you will want to make a note to check for any "non-coding transcripts" and other phrases that I missed and replace them with abbreviations.

Hofmann AH. Scientific writing and communication: papers, proposals, and presentations, 2nd ed. Oxford Univ. Press; 2014.
Schultz DM. Eloquent science. American Meteorological Society; 2009.
Zeiger M. Essentials of writing biomedical research papers, 2nd ed. McGraw-Hill; 2000.

Z komentarzem [AR6]: I'm not sure if this makes sense: wouldn't proteins be present only after translation, which always occurs after transcription, i.e. aren't protein levels are always post-transcriptional? And "protein level of genes" also doesn't seem to make sense, as genes are not proteins. Maybe write "…changes in the levels of proteins coded for by those miRNA"?

(MALAT1), a well-conserved lncRNA that is implicated in diseases in humans, was shown to bind MiR-34a in melanoma cells, thereby lowering miR-34a levels [12]. However, the role of MALAT1 in DILI has not yet been elucidated, and in general, our knowledge concerning the role of lncRNAs in DILI is still limited even in mammals [13].

Successful applications of RNA-Seq technology for resolving problems pertinent to fish biology and immunology prompted us to use RNA based methods to investigate patterns of MC-LR-induced liver injury in a teleost fish, the whitefish *(Coregonus lavaretus)*. Our previous results showed that repeated exposure of whitefish to MC-LR results in severe liver damage, followed by an unexpected resilience to further exposures to the toxin and regeneration of the damaged liver structure [14]. We showed that, in these adaptations, MC-LR regulates several hepatic miRNA signaling pathways and alters the expression profiles of miRNAs over the short-term [15] and long-term [16], suggesting extensive transcriptome rebuilding during these processes. Because lncRNAs have been implicated in species-specific adaptations (e.g. adaptation of zebrafish to cold [17]), a similar adaptation may also have occurred, involving novel lncRNAs.

Biologically active lncRNAs are present in zebrafish [18] and rainbow trout [19]. However, these are species with well-established annotated genomes, and a reference genome is not available for the majority of species. In the absence of an annotated genome, separating NCTs from PCTs requires a more challenging bioinformatic approach. As the foundation for our approach to profile lncRNAs in MC-LR-induced liver injury, we *de novo* assembled a/the whitefish liver transcriptome. Using a step-by-step pipeline designed to filter out redundant contigs, we were able to dissect transcripts without coding potential, which were differentially expressed in whitefish liver after exposure to MC-LR. We identified a list of putative lncRNA transcripts,

Z komentarzem [AR7]: If you mean "…a similar adaptation to MC-LR exposure that involved novel lncRNAs may have occurred in the whitefish." it would be better to write it with those details in the sentence in that order.

Z komentarzem [AR8]: The choice of a/the has to do with how certain you are that your transcriptome is THE only, correct one. Watson and Crick entitled their seminal article on DNA "…A Structure For Deoxyribose Nucleic Acid" because they weren't so sure it was the only possible structure. So, you need to make the choice here based on how confident you are that this is THE only and correct transcriptome.

— **sformatowano:** Wyróżnienie

Z komentarzem [AR9]: "Dissect" means to analyze every tiny detail of the transcripts. It seems strange in this context, and a quick search didn't turn up any papers by native speakers using the word in this context. Perhaps another word like "identify"?

— **sformatowano:** Wyróżnienie

both novel and orthologous to known lncRNAs in other species. In addition, we showed that treatment of fish with MC-LR may affect levels of non-coding transcripts that may be 3' autonomous untranslated regions of mRNAs. Furthermore, by showing how MC-LR changes expression patterns of putative lncRNAs, including MALAT1, we extended knowledge of the underlying mechanisms of liver injury in whitefish. Our findings contribute to better understanding of the role of ncRNA in the molecular response to MC-LR-induced liver injury in fish.

Methods

Fish maintenance and exposure

Fish maintenance and exposure were conducted at the Department of Salmonid Research in Rutki (Inland Fisheries Institute in Olsztyn; Poland). All animal-related procedures were approved by the Local Ethical Commission in Olsztyn, Poland (resolution No. 44/2016 of 30th November 2016). Juvenile whitefish (mean ±standard deviation: 29.9 ±1.6 g, 17.0 ±0.4 cm) were kept in flow-through tanks supplied with well (underground) water. Water temperature in the tanks was 9.3 ±0.2 ºC and oxygen level was 10.3 ±0.6 $mg \cdot L^{-1}$. Throughout the experiment, all the fish were fed with a minimal feeding procedure dependent on the water temperature, caloric content of the feed, and predicted fish mass. However, 1-2 days prior to exposure (intraperitoneal injections) or collection of samples, the fish were deprived of food.

The dose of MC-LR (100 $\mu g \cdot kg^{-1}$ of body mass) and the treatment periods (1, 3, and 9 days) were based on our previous studies on molecular and physiological responses of whitefish to this toxin [14, 16, 21]. MC-LR (purity ≥ 95%; HPLC) was obtained from Enzo Life Sciences (Enzo Biochem, Inc.; USA) and dissolved in phosphate buffer saline (PBS) as a solvent vehicle. Prior to exposure, the fish were anesthetized; they then received an intraperitoneal injection of the MC-LR solution. To maintain continuous exposure, the MC-LR injection (100 $\mu g \cdot kg^{-1}$ of body mass) was repeated after 7 days of the experiment. Fish that received an intraperitoneal injection with pure phosphate buffer saline (PBS) served as a negative control group. Throughout the exposure period, fish from the different groups were kept in separate tanks.

After each exposure period (1, 6, or 9 days), randomly selected individuals from each group were euthanized, and fragments of their livers were collected and preserved in RNAlater solution according to the manufacturer's recommendations (Sigma-Aldrich; Germany).

RNA isolation, sequencing and initial* de novo *assembly

— **sformatowano:** Czcionka: Nie Kursywa, Czcionka o złożonym skrypcie: Nie Kursywa

Total RNA was extracted from the RNAlater-preserved liver fragments (approximately 20 mg) using a mirVana isolation kit (Life Technologies) according to the manufacturer's protocol. RNA integrity was evaluated with an Agilent Bioanalyzer 2100 with an Agilent 6000 Nano Kit. Samples with mean RIN > 8 were taken for library preparation with the Illumina TruSeq Stranded mRNA Library Prep protocol. The libraries were sequenced with an Illumina HiSeq4000 sequencer (250–300 bp insert cDNA size, PE150, 50M reads, 15Gb).

Z komentarzem [AR10]: "With" implies that a person operated/controlled the machine/software/etc. If a person just pushed a button and the device/program did everything, please replace with "by".

Z komentarzem [AR11]: See above.

The workflow used to profile changes in expression of putative ncRNAs in MC-LR induced liver injury in whitefish is shown in Figure 1. Quality control of raw sequencing reads was performed with FastQC, version 0.11.8. To remove adapter sequences and low-quality bases, the reads were

Z komentarzem [AR12]: I think the second-to-last box in Figure 1 might read better as "Combine pairs from the same mRNA strand."

processed using Trimmomatic, version 0.36 [22]. After quality trimming, every 6th read (starting from the 6th), was selected for downstream analysis. Selected reads were assembled into a reference genome using Trinity, version 2.5.1, with the default parameters [23]. Trimmed reads were mapped back to the reference genome using Bowtie2, version 2.3.5.1 [24].

ncRNA identification pipeline

The following pipeline was based on [25]. First, the Trinity *de novo* assembled genome was filtered for redundant transcripts using the cd-hit-est algorithm of CD-HIT [26] with a sequence identity threshold of 0.9. Filtering by expression was executed with RSEM [27] implemented by the Trinity-provided perl script 'align_and_estimate_abundance'. Transcripts with expression levels below FPKM=1.50 were filtered out from the data set.

To assess the quality of the filtered *de novo* assembled transcriptome, we quantified its completeness by comparing it with a set of highly conserved single-copy orthologs. Using the BUSCO (Benchmarking Universal Single-Copy Orthologs) v2 pipeline [28], we compared our assembly with the predefined set of 3640 Actinopterygian single-copy orthologs from the OrthoDB v10 database [29]. BUSCO analysis calculated the number of BUSCOs whose length was within two standard deviations of the mean length of the given BUSCO (complete BUSCOs, C), complete BUSCOs represented by single-copy transcript (single-copy BUSCOs, S), complete BUSCOs evidenced by more than one transcript (duplicated BUSCOs, D), partially recovered BUSCOs (fragmented BUSCOs, F) and not recovered BUSCOs (missing BUSCOs, M). To compare our assembly, we repeated exactly the same procedure with the most complete whitefish whole transcriptome available to date [30].

— **sformatowano:** Wyróżnienie

Z komentarzem [AR13]: Something seems wrong here. You are using BUSCO in two ways. For example: "Benchmarking Universal Single-Copy Orthologs analysis calculated the number of Benchmarking Universal Single-Copy Orthologs-s" (you added a plural -s to a plural -s). After you figure out what to do, you'll need to revise this whole paragraph.

— **sformatowano:** Wyróżnienie

Z komentarzem [AR14]: Did BUSCO also calculate the numbers of these things? If so, change "number" to "numbers" at the beginning of the sentence. Otherwise, there is some other problem in the sentence.

Z komentarzem [AR15]: Compare it with what? If "compare" is the right word here, rewrite as "To compare our assembly with XXX, …". Or maybe you want to say "'verify' our assembly", "'assess' the quality of our assembly", or something like that?

— **sformatowano:** Wyróżnienie

Next, the transcripts were searched for open reading frames (ORFs) by Transdecoder v2.0.1 [31]. To identify protein coding transcripts (PCTs), ORFs and transcripts were searched against the UniProt-Swiss-Prot and Atlantic salmon proteins reference databases (GCF_000233375.1) using blastp and blastx from the BLAST+ suite with a threshold E-value of 1×10^{-3} [32]. Protein family searches were performed with the Pfam 32.0 database [33] using the ORF protein sequences in HMMER 3.2.1. Finally, the top BLAST hit based on the bit score, E-value and percent alignment, and all HMMER hits were loaded into Trinotate 3.2.1 to generate an annotation report [34]. Based on the report, transcripts that were not PCTs were then filtered against the RFAM database, v12.0 [35] by the cmscan algorithm implemented by Infernal, version 1.1.3 [36]. Any hit that Infernal considered significant using the default parameters was filtered out (and labeled as a known ncRNA). All remaining putative novel NCTs were further validated by calculating coding potential using CPC [37].

— **sformatowano:** Wyróżnienie

Z komentarzem [AR16]: For simplicity, maybe it would be better not to use the word "version", e.g. "Infernal 1.1.3". In any case, you should be consistent because sometimes you write "version" and sometimes you don't. Pick one style or the other and use it throughout the text.

— **sformatowano:** Wyróżnienie

To further verify that the remaining contigs were completely separated from mRNAs, putative novel non-coding transcripts were subjected to a blastn search against the Atlantic salmon Reference RNA Sequences database (NCBI, refseq_rna). Any transcript that was identified as "mRNA" was set together in a pair with the corresponding PCT of the same mRNA. Only transcripts which had a corresponding PCT were subjected to further analysis (putative autonomous 3'UTR).

Z komentarzem [AR17]: If this refers to the transcripts which had a corresponding PCT, it should be plural (3'UTRs).

At this point, all transcripts that were considered to be either putative known RNAs or novel non-coding RNAs, as well as transcripts identified as Atlantic salmon proteins and putative autonomous 3'UTR transcripts, were counted in each sequenced sample using samtools idxstats (Figure 1B) [38].

Z komentarzem [AR18]: "Putative" means "generally considered to be", which doesn't make sense if the transcript is "known". Either it is known, or it is putative, but it doesn't seem possible to be both.

Free Energy Levels of Non-Coding Transcripts

The minimum free energy of each transcript was calculated using the rnafold algorithm implemented by ViennaRNA 2.4.12 [39] using the following options: -p –d2 –noLP. The minimum free energies of the transcripts were then compared to the minimum free energy of a randomly selected set of protein coding transcripts.

Z komentarzem [AR19]: One "energy" (like a mean or median free energy) or the individual "energies" of all the transcripts?

Functional annotation of putative autonomous 3'UTRs and PCTs of the same mRNA

GO analysis (http://www.geneontology.org) was performed to construct gene annotations. To retrieve GO IDs for particular proteins, we used the Retrieve/ID mapping tool from the UniProt website [40]. WEGO (Web Gene Ontology Annotation Plot) was used to visualize the results [41]. A P-value < 0.05 was considered to indicate a statistically significant difference.

qPCR

To profile putative lncRNA expression, reverse transcription (RT) was carried out using SuperScript IV Reverse Transcriptase (Thermo Scientific; USA). The reaction contained 1 µg total RNA, 4 µL 5× RT buffer, 1 µL 0.1 M DTT, 1 µL 10 mM dNTP mix, 1 µL Ribonuclease Inhibitor and SuperScript IV RT enzyme, and 1 µL 50 µM Oligo(dT)$_{20}$ primer. The reaction was carried out at 23°C for 10 min, 55°C for 10 min, and 80°C for 10 min. Synthesized cDNA samples were diluted (20×), stored at −80°C, and thawed only once, just before the amplification.

Real-time PCR was used to determine levels of putative lncRNAs in the cDNA samples. Reactions were carried out in final volumes of 20 µL, consisting of 10 µL Power SYBR Green

Z komentarzem [AR20]: "the relative expression" is probably a better phrase, as you were normalizing to a reference gene, and thus did not perform any kind of absolute quantification.

PCR Master Mix (Life Technologies, USA), 0.25 µM of each primer (forward and reverse; Supplementary Table 2), 1 µL cDNA template and 7 µL PCR-grade water. Amplification was performed with a Quant Studio 5 Real-time PCR System (Applied Biosystems; USA) with the following conditions: 95°C for 10 min, then 45 cycles of 95°C for 15 s and 60°C for 1 min. The reaction for each sample was carried out in duplicate. No-template controls (NTCs) were included to test for the possibility of cross-contamination. To check the quality of each PCR product, melting curve analyses were performed after each run. Data from the treated group (MC-LR) were normalized to XXX as a reference gene, and compared to the control group (PBS).

Statistical methods

Contigs that were differentially expressed (DE known and novel lncRNAs, DE 3'UTRs, DE PCTs) in the MC-LR-treated and the control (PBS) groups were indicated using the DESeq2 package (version 1.28.0) [42] for R (version XXX) [XXX]. Adjusted p-values were calculated with Benjamini and Hochberg's method using the default settings to maintain a nominal false discovery rate of 0.1 [43]. Thresholds of a $\log_2$ fold-change > |2| and an adjusted p-value < 0.001 were used to filter out contigs with the smallest and least statistically significant differences between groups.

Before assessing the differences in minimum free energy and content of GC base pairs, we used histograms and normal Q-Q plots (shown in Additional File 1) to assess the distribution of the data. These methods indicated that, considering the large sample size, any deviations from normality were too small to be important. Thus, we assessed differences between groups using

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Z komentarzem [AR21]: It looks like you were using the ddCt method or one of its variants (e.g. an efficiency corrected method), so the data from both groups was normalized to the same reference gene(s) before the groups were compared. Normalization is the one of the deltas ("d"), and comparison between groups is the other one.

If this was indeed a ddCt-type analysis, then rewrite as follows: "For normalization of data from the treatment (MC-LR) and control (PBS) groups, XXX was used as a reference gene."

Z komentarzem [AR22]: You should give the version for R and cite the R Core Team when using the software. Type "citation()" in the R console for the necessary information.

Z komentarzem [AR23]: You capitalized "p-value" above (i.e. "P-value"). Check the guidelines for authors, and if they don't have a preference, pick one or the other and use it consistently.

Z komentarzem [AR24]: I'm not familiar with DESeq2 software, but in Benjamini and Hochberg's (1995) procedure, using a threshold of an adjusted p-value of 0.1 gives an FDR of 0.1.

Benjamini Y, Hochberg Y. Controlling the false discovery rate: a practical and powerful approach to multiple testing. Journal of the Royal statistical society: series B (Methodological). 1995 Jan;57(1):289-300.

— **sformatowano:** Indeks dolny

Z komentarzem [AR25]: A log2 fold change > |2| is a difference greater than four-fold, e.g. 4.1 to 1 or more. This seems like a large difference, and I have seen papers by experts on gene expression and bioinformatics that consider a fold change of 1.5 ($\log_2$ fold-change of 0.6) to be biologically relevant. Therefore, consider changing to "…smaller and less statistically significant differences…".

— **sformatowano:** Nie Wyróżnienie

Welch's t-test, which is robust to violations of the assumption of homoscedasticity. 95% confidence intervals for differences are shown in square brackets in the main text, e.g., [9.5, 11.7]. Statistical calculations were handled in Python's statistical modules (Sci-Py, StatsModels). Confidence intervals were calculated using the R Tidyverse package.

Statistical calculations for qPCR data were performed using GenEx 7.0. To assess the significance of difference between groups, ANOVA was used, followed by Dunnett's *post hoc* test.

Data availability

The data from this study have been submitted to the NCBI SRA database (accession #xxx). The accession numbers for data from the individual samples, read numbers and concentrations of total RNA in the extracts are given in Additional File 2.

Results

Sequencing results

The number of raw reads in samples ranged from 50797688 to 70885836. The effective rate ranged from 95.88 to 99.25% ([Clean reads/Raw reads]*100%), with a stable base error rate at 0.03% in all samples. Content of GC base pairs ranged from 47.23 to 50.50%. Detailed statistics on the quality of sequencing data for each sample are presented in Additional File 2.

Z komentarzem [AR26]: It's unclear which statistical calculations you are referring to here. Please rewrite this and mention the calculations by name.

Z komentarzem [AR27]: Tidyverse is a collection of packages, not a package. Although I have probably referred to tidyverse in a paper myself, it's definitely better to refer to the specific package that calculated the confidence interval—and the people who made this free software would appreciate a citation too, if you can fit it into your reference list. Type citation("name_of_package") in the R console for the necessary details.

Z komentarzem [AR28]: First, hopefully you log-transformed the qPCR data before using ANOVA and Dunnett's test. This is a crucial step in qPCR data analysis, and should be mentioned here. See, for example:
Rieu I, Powers SJ. Real-time quantitative RT-PCR: design, calculations, and statistics. The Plant Cell. 2009 Apr 1;21(4):1031-3.
Beal J. Biochemical complexity drives log-normal variation in genetic expression. Engineering Biology. 2017 Jul 27;1(1):55-60.
Therefore, you could write "…ANOVA was used after log-transforming the data…".

Second, the variant of ANOVA should be specified. (I assume it was "one-way independent ANOVA", which you can write here, if correct.)

Finally, I took the liberty of removing "analysis of variance". The abbreviation ANOVA is almost as well-known as DNA, RNA or kg, and thus doesn't need to be defined.

Assessing the quality of the de novo assembled liver transcriptome

The number of detected transcripts in our raw *de novo* assembled liver transcriptome was 1,136,890 with an average length of 367 base pairs. After the assembly, we mapped the fraction of trimmed reads back to the assembled liver transcriptome. The fraction of aligned reads was between 97 and 98% per sample. The final assembly, which was used in further analyses, was obtained by filtering too short, redundant and lowly expressed transcripts. The number of transcripts in our final assembly was 420,280 transcripts, with an average length of 594 base pairs. The detailed statistics of the final assembly are presented in Additional File 3.

The BUSCO analysis pipeline revealed that, of the 3640 Actinopterygian single-copy orthologs searched, our final assembly completely recovered 74.9% and partially recovered 7.5% (Table 1). 17.6% of the single-copy orthologs were reported missing from our liver transcriptome.

De novo *transcriptome assembly allowed discovery and classification of non-coding RNAs in whitefish exposed to MC-LR*

Using the procedure for identifying non-coding RNAs in non-model species first described by Harris et al., we managed to first separate PCTs (148,646 contigs) and then discover various long non-coding transcripts in the whitefish (209,270) [25]. Subsequent filtering steps allowed us to separate these transcripts into three non-overlapping groups. The first group contained lncRNAs that showed homology to sequences deposited in the Rfam database (the group of known long non-coding transcripts: 20,272 contigs). The second contained long non-coding transcripts that had homology with non-coding 3' untranslated regions of mRNA sequences deposited in the RefSeq database and had associated PCTs (autonomous 3'UTR/PCT group: 104,024 contigs).

Z komentarzem [AR29]: "The" means that the reader should already know which fraction of trimmed reads, which is confusing to me. I wonder if this sentence would be better simply as "After the assembly, we mapped the trimmed reads back to the assembled liver transcriptome."

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Z komentarzem [AR30]: Maybe "filtering out", meaning that you removed those short, redundant and lowly expressed transcripts?

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— **sformatowano:** Nie Wyróżnienie

Z komentarzem [AR31]: Is there any difference in meaning between this and lncRNA? If not, disregard Polish style guidelines and write in English for an international audience: repeat key words early and often, and always use abbreviations after you have defined them.

See, for example:
Hofmann AH. Scientific writing and communication: papers, proposals, and presentations, 2nd ed. Oxford Univ. Press; 2014.
Zeiger M. Essentials of writing biomedical research papers, 2nd ed. McGraw-Hill; 2000.

— **sformatowano:** Wyróżnienie

— **sformatowano:** Wyróżnienie

The third contained long non-coding transcripts with no homology to any tested database (putative novel long non-coding RNAs: 84,974 contigs). Our filtering process is summarized in the workflow diagram (Figure 1B).

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MC-LR exposure altered expression of evolutionary conserved lncRNAs

To obtain the list of evolutionary conserved lncRNA, transcripts left after removing PCTs were checked for coding potential with the SVM-based Coding Potential Calculator. Non-coding transcripts were further compared with sequences of transcripts deposited in the Rfam database. This produced a list of 20,272 contigs, which were counted and analyzed for differential expression (Figure 2A-D). Among 4,238 differentially expressed (DE) evolutionary conserved non-coding RNAs, there were 94 known, putatively conserved DE lncRNAs identified, including MALAT1, HOTTIP, HOTAIR and HULC (Figure 2E-H). To show similarities in sequence conservation between whitefish and 17 other species, including human, we aligned the sequence of our putative MALAT1 transcript with seed sequences of MALAT1 deposited in the Rfam database (Additional File 4).

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Coexpression of autonomous 3'UTRs and their associated PCTs after MC-LR exposure

To investigate the effects of MC-LR exposure on the expression of non-coding contigs identified as autonomous 3'UTRs, we first set them together with transcripts identified as coding transcripts (PCTs), that 3'UTR is part of. Then, using the DESeq2 package from Bioconductor, we separately calculated the differential expression of the putative autonomous 3'UTRs and their associated PCTs. Finally, we compared the percentages of the corresponding contigs that were

Z komentarzem [AR32]: Something is not right here. Maybe "…, we first paired them with the PCTs that the 3'UTR is attached to."
You already defined and used PCT multiple times, so do not define it again here. You might redefine an abbreviation the first time you use it in the results if, for example, you used it in the Introduction, but then did not use it during the entire Methods section.

both up- or down-regulated and those that were regulated in contrasting directions (Figure 3A). We found that, in the majority of cases, if a putative autonomous 3'UTR was DE, the associated PCT was also DE in the same direction. After 1d of the experiment, over 82% of the transcripts pairs were upregulated and only 11% downregulated. In contrast, after 6d and 9d, up- and down-regulated pairs were present about in the same proportions (around 40%). Moreover, we checked whether the pattern of changes in expression of coexpressed 3'UTR/PCT pairs was reflected in the pattern of changes in expression of all PCT (paired with 3'UTRs and unpaired). We found that at 6d and 9d time points both expression patterns were similar but not identical, however the ratio of up- and downregulated transcripts at 1d totaled 66 and 44% accordingly (Fig. 4).

Gene Ontology analysis indicated similarities in the terms of pairs that were simultaneously upregulated or downregulated after 6 and 9d of MC-LR exposure (Additional File 6). In contrast, upregulated pairs at 1d were enriched in transcription regulator activity and DNA-binding transcription factor activity transcripts when compared with upregulated pairs at 6 and 9d (Fig. 5). Downregulated pairs at 1d were depleted in transcripts involved in enzyme regulator activity processes when compared with downregulated pairs at 6 and 9d of exposure.

MC-LR produced opposing expression profiles of some autonomous 3'UTRs and their associated PCTs

In addition, we found that MC-LR exposure caused contrasting changes in the expression of some 3'UTRs and their associated PCTs (Fig. 3A, green and brown bars). The proportion of these transcripts (7%) was lowest after 1d of MC-LR exposure, and higher at 6 and 9d (19% and 16% accordingly). On the other hand, the proportions of both groups with the opposite

Z komentarzem [AR33]: Is this long description of Methods really necessary here, in the Results sections? One sentence or two is generally okay, so consider cutting everything that is obvious or that the reviewer/reader does not need to understand the results.

Z komentarzem [AR34]: To me, little or nothing is lost by starting the paragraph here.

— sformatowano: Wyróżnienie

Z komentarzem [AR35]: It's difficult to understand this section and Figure 4 (which I understand is unfinished). What you write here does not seem to match the figure at all, so I was unable to make any corrections.

One thing I find confusing is that, if the number of upregulated coexpressed PCTs increases, won't the number of all PCTs (coexpressed or not) also increase, unless the number of independently expressed PCTs decreases? Think about it this way: if the number of male PhD students in your department increases, doesn't the total number of PhD students also increase, unless a female student leaves?

Think about this as you revise this section and the figure—maybe you just need to add a detail or explain something better to make it clearer.

Z komentarzem [AR36]: I think you mean file 5.

Z komentarzem [AR37]: What's the base of the logarithm in figure 5? I think $\log_{10}$ is most common in bioinformatics, but I've also seen $\log_2$. This makes a big difference in how we interpret the strength of the evidence shown by the figure.

Z komentarzem [AR38]: Is this figure readable by a color-blind person? If your reviewer is green-brown colorblind, he may be unhappy. See https://colorbrewer2.org/#type=sequential&scheme=BuGn&n=3 for color-blind friendly palettes of colors.

expression profiles remained at a similar level throughout all the days of exposure. In the group with 3'UTRs downregulated and PCTs upregulated, gene ontology terms again showed similarities on days 6 and 9, whereas on day 1, the transcripts were enriched in terms like 'signaling' or 'response to stimulus'. In contrast, in the group with 3'UTRs upregulated and PTCs downregulated, transcripts involved in 'cell', 'cell part', 'membrane' and 'membrane part' were enriched on days 6 and 9.

Putative novel lncRNAs and PCTs differed in terms minimum free energy, GC base-pair content and length

To determine if the novel whitefish lncRNA candidates differed from PCTs in terms of minimum free-energy (MFE), we used the RNAfold algorithm from the ViennaRNA package. The free energy values of the secondary structures were corrected for the lengths of the sequences (Fig. 6A). The mean length-corrected MFE of the putative lncRNAs was significantly higher than that of the annotated protein coding transcripts (-0.237 kcal/mol/nt ±0.038758 vs. -0.289 kcal/mol/nt ±0.059464 kcal/mol/nt, respectively, $t() = 46.65$, $p < 0.001$, [0.04963723, 0.05399174]). The mean length-corrected MFE of the putative lncRNAs was 1.22 kcal/mol/nt higher than that of the protein-coding transcripts. Moreover, the mean content of GC base pairs was significantly higher in protein-coding transcripts ($t() = 56.16$, $p < 0.001$, [0.06915164, 0.06448719]) (Fig. 6B). Finally, the length of the transcripts was different in both groups (Fig. 6C). In summary, structural differences between the putative lncRNAs and the annotated PCTs validated our methodology for discovery of lncRNAs in whitefish.

Z komentarzem [AR39]: Something is unclear here. As I understand it, you said that the proportion of transcripts with opposing expression profiles changed, and now you say it was similar.

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Z komentarzem [AR40]: "comparatively enriched", i.e. compared to days 6 and 9?

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Z komentarzem [AR41]: This whole set of comparisons seems unbalanced and incomplete. You say what transcripts were enriched on day 1 in the group with 3'UTRs downregulated and PCTs upregulated, but you don't say anything about the kinds of transcripts on days 6 and 9. Then in the group with 3'UTRs upregulated and PTCs downregulated, you give details about days 6 and 9, but not day 1. It's like a table with blank spaces.

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Z komentarzem [AR42]: t-statistics are usually rounded to two decimal places, so I took the liberty of doing that for you. Also, you should put the degrees of freedom (often labeled "df" in R and Python output) inside the brackets, i.e. the "(…)".

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Z komentarzem [AR43]: These two sentences appear to say different things. The second sentence has a larger difference that does not match Figure 6A.

Z komentarzem [AR44]: add degrees of freedom

Z komentarzem [AR45]: This is figure 6C. You should correct the figure caption too.

Z komentarzem [AR46]: In fact, the most common length in both groups looks very similar in Figure 6B—look at the peaks of the two distributions. I would suggest saying that "the distribution of transcript lengths differed between the putative lncRNAs and the annotated PCTs," and maybe adding "with more PCTs with longer sequence lengths."

MC-LR altered the expression profiles of the identified putative novel lncRNAs

Using the DESeq2 package of Bioconductor, we investigated whether MC-LR induced changes in the expression profiles of the putative novel lncRNAs discovered in this study. Using an adjusted p-value of 0.001 and a 2-fold expression as cutoffs, we identified 1739 and 2690 transcripts that were either up- or down-regulated between days of exposure, respectively. Figure 7 shows volcano plots of up- and downregulated putative novel lncRNAs (Fig. 7A-C) and Venn diagrams with number of transcripts specific to each time period, as well as those which overlap between days of exposure to MC-LR (Fig. 7D, E). 31% of all up- and 34.8% of all down-regulated lncRNA transcripts were common between days 6 and 9 showing similar expression changes between these time periods. On the other hand, the similarities of common transcripts between 1d and 6d and 9d ranged from 0.23% to 5.31%. The same pattern could be also observed in the protein coding transcripts identified in this study (Fig. 4).

qPCR confirmed aberrant expression of selected transcripts

To validate the RNA-Seq data, we selected three known lncRNAs (Fig. 8) and 10 putative novel lncRNAs (five upregulated and five downregulated, Additional File 6) and designed a qPCR study that re-analyzed their levels. The qPCR data indicated statistically significant changes in the expression of all selected transcripts except MALAT1 transcripts. Our RNA-Seq results show that MALAT1 was present at a high levels in normal whitefish liver and its expression was downregulated at 6 and 9d after MC-LR exposure. High levels of MALAT1 in normal liver was further confirmed using qPCR, however downregulated expression was not confirmed by Dunnett's *post hoc* test (Fig. 8A). This could be caused by high variance of Ct values due to too low abundance of transcripts for qPCR, which was close to the method's detection level.

Z komentarzem [AR47]: A $\log_2$ fold-change of 2 is a regular fold-change of 4. Check what you did and correct throughout the manuscript as appropriate. Here you should write either "...and a fold-change of 2…" or "…and a $\log_2$ fold-change of 2…"

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Z komentarzem [AR48]: This means you were comparing the MC-LR group on day one to the MC-LR group on days six and nine. If you were comparing the MC-LR group to the control group on each day, write "…up- or down-regulated by MC-LR exposure."

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Z komentarzem [AR49]: In the figure, 319 + 935 = 1,254 transcripts are downregulated on both day 6 and day 9. That's 46.6% (1,254/2,690) of all downregulated transcripts. I think maybe you meant to write something like "In terms of changes in the expression of the identified transcripts, days 6 and day 9 were more similar to each other than to day 1. 31.0% and 34.8% of all up- and downregulated lncRNA transcripts, respectively, were downregulated on both day 6 and day 9, but not on day 1 of the exposure (Fig. 7D, E)."

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Z komentarzem [AR50]: If you're wondering about "three" and "10", in APA style (which is accepted by most journals) you write out the word when it's a number less than 10 and you are talking about things. If you are writing about SI units like kg, days, etc., you write the numeral, i.e. 3 days.

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Z komentarzem [AR51]: This seems like it belongs in the discussion.

— **sformatowano:** Wyróżnienie

Maciek,

I think you need to address two things in the "qPCR confirmed aberrant expression of selected transcripts" subsection. First, you say above that MALAT1 is abundant and present at high levels, but then you say its abundance is too low for reliable qPCR. That seems contradictory—do you need to explain more or change something? Also, in the figure, the variance of MALAT1 looks similar to that of the other transcripts.

Second, did you log-transform the data before analysis? If not, that could be the cause of the "non-significant" difference in MALAT1 because without a log transformation you have very little statistical power to detect differences.

If you did log-transform the results, or if the difference is still non-significant after transformation, I would emphasize that your qPCR results are consistent with your RNA-Seq results, just not conclusive. Remember that the correct interpretation of a statistically non-significant difference is "inconclusive", not "no difference". Amrhein et al. (2017) write this:

'The "most devastating" of all false beliefs is probably that "if a difference or relation is not statistically significant, then it is zero, or at least so small that it can safely be considered to be zero" (Schmidt, 1996). For example, if two studies are called conflicting or inconsistent because one is significant and the other is not, it may be implicitly assumed that the nonsignificant effect size was zero (Cumming, 2012, p. 31).' You seem to be close to doing this here: interpreting the qPCR data as potentially conflicting with the RNA-Seq data, when in fact it is consistent. Both methods show downregulation of MALAT1 on days 6 and 9. The only thing is that the qPCR data is not strong enough to be statistically conclusive.

I understand that you may be worried that some reviewers don't understand statistics and may attack this interpretation. Don't worry, you can easily defend this interpretation by citing the American Statistical Association's statement on p-values (Wasserstein and Lazar 2016) or any good textbook on statistics, like Motulsky (2018)—

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Maciek Woźny has a copy of that book. I don't think any reviewer can successfully argue with the American Statistical Association, a textbook, and a third citation (i.e. Amrhein et al. 2017).

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You can write the Results like this, emphasizing the size of the difference and not the lack of statistical significance: "Although the difference did not meet our predetermined criterion for statistical significance (P = 0.XXX, Dunnett's test), MALAT1 expression was X.X-fold lower 6 and 9d after MC-LR exposure than in the control group. The other nine transcripts were all significantly DE (P < 0.05)."

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I would suggest starting to discuss MALAT1 like this (in the Discussion): "When assessing the qPCR validation results, it is useful to remember that the American Statistical Association and other experts have stated that non-significant results are correctly interpreted as inconclusive, not as indicating no difference (Wasserstein and Lazar 2016, Amrhein et al. 2017, Motulsky 2018). Thus, although the qPCR data on MALAT1 was not statistically conclusive, it was consistent with the RNA-Seq data, in that both techniques suggested or indicated, respectively, downregulation of this transcript 6 and 9d after MC-LR exposure.

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Amrhein V, Korner-Nievergelt F, Roth T. The earth is flat (p > 0.05): significance thresholds and the crisis of unreplicable research. PeerJ. 2017 Jul 7;5:e3544.

Motulsky H. Intuitive biostatistics: a nonmathematical guide to statistical thinking. Oxford University Press, USA; 2018.

Ronald L. Wasserstein & Nicole A. Lazar (2016) The ASA Statement on *p*-Values: Context, Process, and Purpose, The American Statistician, 70:2, 129-133, DOI: 10.1080/00031305.2016.1154108

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Discussion

In this study, using RNA-Seq data, we identified a list of putative lncRNA transcripts involved in MC-LR induced liver injury in whitefish, a non-model species without a reference genome. Further qPCR validation of selected putative lncRNA candidates confirmed the participation of these transcripts in MC-LR induced liver injury in whitefish. We showed that the altered expression profiles of lncRNAs could serve as a potential biomarkers of liver injury in whitefish.

The lack of standardized methodologies for discovery of lncRNAs poses a challenge for the analysis and interpretation of RNA sequencing data. The outcome of pipelines designed to discover lncRNAs in RNA-Seq data strongly depends on factors which precede *in silico* analysis, such as RNA isolation or the method of preparing sequencing libraries [44]. Therefore, methods designed for enriching polyadenylated protein-coding mRNAs may not be optimal for recovering lncRNAs that are present at low levels. On the other hand, designing a large-scale RNA-Seq experiment with a large set of samples is demanding and usually some trade-offs must be made [45]. Because the main aim of our RNA-Seq experiment was to dissect the profiles of protein-coding genes, Illumina's TruSeq Stranded mRNA protocol was chosen (Brzuzan et al., in preparation). However, this protocol can also be used for lncRNA discovery [44]. In fact, the majority of biologically functional lncRNAs reported to date are polyadenylated [46], thus approaches based on enriching polyadenylated transcripts to discover functional lncRNAs are common in pipelines based on annotated genomes, as well as those based on *de novo* assembled transcriptomes. For example, 54,503 putative lncRNAs were discovered in rainbow trout [19]

Z komentarzem [AR52]: "analyze"? (see note on "dissect" above)

Z komentarzem [AR53]: Confusing—if this was the main aim, you should change the title to reflect this.

and 122,969 putative lncRNAs were reported in *Rhinella arenarum* [47]. Our pipeline based on the *de novo* assembly of the liver transcriptome allowed us to obtain 209,270 non-coding transcripts longer than 200 nt, of which 84,974 were labeled as putative novel lncRNAs(see further discussion).

Difficulties in comparing results of RNA-Seq *in silico* analysis based on *de novo* assembled transcriptomes may be attributed to multiple factors. For example, poor reproducibility of *de novo* based analyses [48] could come from the source of the reference genome, in addition to the methodologies used in preparation of the sequencing libraries (i.e. selection of tissues). Last but not least, to obtain reliable and comparable results in discovery of lncRNAs, sequencing depth should be considered. It is estimated that, in human samples, >200 million paired-end reads are required to detect the full range of transcripts, including all possible isoforms [49]. However, this number can be much lower for differential expression analyses. For example, if the expectation is that the expression of abundant transcripts changes across conditions, 36 million reads per sample may be sufficient [45]. Because it was expected that (i) MC-LR will drastically change expression profiles of transcripts [16] and (ii) polyadenylated lncRNAs are expressed at higher abundances than non-polyadenylated lncRNAs [50], we sequenced our liver samples with 50 million reads per sample, which was sufficient to identify evolutionary-conserved and novel transcripts.

In organisms without a conclusively proven reference genome, good quality *de novo* assembled transcripts are a prerequisite for obtaining meaningful results in downstream analysis, such as discovery of novel transcripts [51]. Here, we assembled a whitefish liver transcriptome from 52 liver samples that originated from different experimental groups, including some that were part

Z komentarzem [AR55]: "Here," means in this study, but you cite another work. Maybe "Combining data from this study and another,…"

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of another study, which extended the scope of available transcripts used for the assembly (Brzuzan et al., in preparation, Additional File 2). BUSCO analysis, which estimates assembly quality based on evolutionary-informed expectations of gene content from orthologues selected from OrthoDB, showed 74.9% completeness of Actinopterygian core genes (OrthoDB v10). For comparison, the current best assembly of a whitefish full-length transcriptome based on a whole fish homogenate showed only slightly higher completeness (76.6%) (OrthoDB v10, Table 1) [30]. Importantly, BUSCO recovery estimates tend to be higher in full organism assemblies than in those assembled from a set of separate tissues. For example, a whitefish tissue-based full-length transcriptome deposited in the PhyloFish database showed only 26% completeness [30]. Because our assembly of a whitefish liver transcriptome showed completeness similar to the current best whitefish whole transcriptome assemblies, we believe that it not only provided a solid foundation for our analysis, but it could also extend the completeness of current and future assemblies of whitefish transcriptomes.

Designing an accurate step-by-step pipeline for discovery of lncRNAs is an essential step for producing high-quality results. This is even more important for non-model species without a reference genome, as redundancy tends to be higher in those analyses. As a consensus state-of-the-art integrated pipeline does not yet exist, we decided to base our pipeline on a previously described pipeline for identification of lncRNAs in non-model species [25]. The core aim of this pipeline was to remove known PCTs and other non-coding RNAs in a sequence of filtration steps. As a result, the pipeline predicts novel lncRNAs, then uses software that employs support vector machines in an attempt to validate the novel transcripts by calculating their coding probability. Unfortunately, as both the pipeline and the validation process use BLAST results

Z komentarzem [AR56]: Other than what? Maybe you want to say "…and known NCTs…" (or "ncRNAs")?

Z komentarzem [AR57]: Support vector machines classify data—while the software may calculate coding probability, I don't think the support vector machines were directly responsible for this. See, for example, https://en.wikipedia.org/wiki/Support_vector_machine https://towardsdatascience.com/support-vector-machine-introduction-to-machine-learning-algorithms-934a444fca47

Maybe just remove the information about SVMs from the sentence?

with varying levels of confidence, this validation is in fact only pseudo-independent, and thus the presented transcripts are predictions at best, and only experimental evidence can validate their true function [25]. However, a lack of an assessment of the sensitivity or the specificity of a pipeline is common among current studies aiming to classify novel lncRNA transcripts, particularly in non-model species without well-annotated genomes. Here, we show that the putative lncRNAs differ substantially from the PCTs in terms of minimum length-corrected free energy, GC content and length (Fig. 6). These factors are considered crucial for RNA stability and are in line with previous reports of lncRNAs having different stability as compared to PCTs in fish [19]. Additionally, to validate the results of our pipeline, we performed a qPCR validation of selected putative lncRNA transcripts (Additional File 6).

Based on the similarity of our transcripts to the non-coding sequences deposited in the Rfam database, we identified putative whitefish liver lncRNAs whose expression was changed after MC-LR administration (known differentially-expressed lncRNAs). We found that MC-LR altered expression of potent regulators of genes, such as HOTAIR, HOTTIP, HULC or MALAT1 (Fig. 2E-H). Generally, lncRNAs are poorly conserved among species [8], but some of them, for example MALAT1, are conserved in mammals [52] as well as in other vertebrates, such as zebrafish [53], suggesting that they have important conserved biological roles. Moreover, MALAT1 is one of the most abundantly expressed lncRNAs in normal tissues [54, 55], even similar in this regard to many protein-coding genes, such as GAPDH [56]. MALAT1 expression has been shown to be either upregulated (lung cancer, hepatocellular carcinoma) or downregulated (colorectal cancer, breast cancer) [55], indicating its role is either cancer-promoting or tumor-suppressing. At the functional level, MALAT1 has been shown to bind

Z komentarzem [AR58]: Or “differ markedly”, or just “differ”? See my notes above on the use of markedly/substantially..

Z komentarzem [AR59]: Not sure if a reviewer will accept this; “distribution of transcript lengths” is probably safer. See my notes in the Results section for more on this.

Z komentarzem [AR60]: As I understand it, minimum free energy mostly has to do with the stability of the secondary structure, i.e. the shape of the transcript after it folds, not the stability of the transcript itself, i.e. how quickly/easily the transcript falls apart. Maybe “…crucial for RNA secondary-structure stability, and our results are in line with previous reports from studies of fish, in which lncRNAs differed from PCTs in the same manner”?

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Z komentarzem [AR61]: If you are going to say in the Results that “downregulated expression [of MALAT1] was not confirmed by Dunnett’s *post hoc* test”, then I expect the reviewers will give you problems about this paragraph and the next two. Indeed, if I didn’t have the statistical background to correctly interpret your qPCR results, I would be skeptical and confused here.

I will focus on correcting language, not restructuring the text, but I have several quick suggestions: First, use my notes on MALAT1 in the Results to start this section of the discussion. Second, continue by writing something like “Thus, on the basis of the RNA-Seq and qPCR data, we conclude that MC-LR altered expression of potent regulators of genes, such as HOTAIR, HOTTIP, HULC, and most likely altered expression of MALAT1 as well.” Then continue discussing MALAT1.

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Z komentarzem [AR62]: In the Results you say its expression is too low for reliable quantification—that needs to be clarified or removed.

various miRNAs, which either promoted [57] or decreased cancer progression [58]. For example, knocking down MALAT1 in melanoma cells significantly upregulated the expression of tumor suppressing miR34a [59]. Interestingly, our previous results demonstrated that miR34a was upregulated in whitefish liver after MC-LR exposure [60]. Because our RNA-Seq and qPCR results showed that MALAT1 was present at a high level in normal whitefish liver and its expression was downregulated at 6 and 9d after MC-LR exposure, this could indicate that decreased abundance of MALAT1 may also be linked with upregulated levels of miR34a in whitefish liver after MC-LR exposure. Although it is too soon to speculate on whether and how this effect could potentially underlie the molecular mechanisms of MC-LR-induced liver injury, this finding adds to the little that is known about possible lncRNA and miRNA interactions in liver cells.

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Z komentarzem [AR63]: See my previous comments on MALAT1 qPCR results.

Previous research on MALAT1 variability has demonstrated that the majority of its transcripts are not polyadenylated [56]. However, it has been revealed that, in mammals, MALAT1 not only has a highly abundant short isoform that lacks a poly(A) tail, but also has a long isoform that is present at a much lower level and has a genomically encoded poly(A) tract. The longer isoform is further processed by RNAse P and RNAse Z to the shorter isoform. In this study, using a library preparation method that enriched only polyadenylated transcripts, we might have detected mainly polyadenylated MALAT1 transcripts, thus confirming the presence of that isoform in fish. To better study the biological function of these lncRNA, further research is required to investigate both the structural diversity and gene regulation of MALAT1.

Previous research on MALAT1 variability demonstrated that the majority of MALAT1 transcripts are not polyadenylated [61]. As we were using a library preparation method that enriched only polyadenylated transcripts, it was not possible for us to detect non-polyadenylated MALAT1. However, it has been revealed that, in mammals, MALAT1 has two isoforms: a highly abundant short isoform, which lacks a poly(A) tail, and a long isoform that is present at a much lower level, but has genomically encoded poly(A) tract. The longer isoform is further processed by RNAse P and RNAse Z to the shorter isoform. In our pipeline we might have detected the polyadenylated isoform of MALAT1. Unfortunately, this limits discussion of the biological role of MALAT1 in MC-LR-induced liver injury, but should be addressed in further studies.

Z komentarzem [AR64]: This seems to be mostly a repetition of what was written above.

After removal of PCTs and known NCTs, a large group of the remaining NCTs still mapped to the mRNAs deposited in the Reference Sequence database (RefSeq). However, after closer examination of particular BLAST hits, we noticed that the vast majority of our remaining NCTs mapped not to the coding parts of matched mRNA sequences, but to their non-coding 3'UTR regions. The presence of autonomous 3′UTR transcripts separated from their associated mRNAs has been documented in studies on mouse and human cells [62–64]. Because 3'UTR regions that were considered to be a part of the canonical transcripts are in fact biologically significant autonomous units participating in post-transcriptional regulation [64], we analyzed how expression of our putative autonomous 3'UTR transcripts corresponded to that of the PCT from the same mRNA. We found that, in the normal (unchallenged) condition, almost the same number of mRNAs had a significantly higher number of 3'UTR transcripts (48% of differentially expressed (DE) mRNAs) as those which had a higher number of PCTs (52% of DE mRNAs). This may indicate that, in normal whitefish liver, expression of those transcripts remains in a stable relationship. Moreover, we showed that exposure to MC-LR increased the abundance of PCTs while decreasing that of 3'UTR transcripts from the same mRNA (Fig. 3B). In pairs in which expression was changed after MC-LR exposure, 60% of the pairs had more PCTs than 3'UTR transcripts. In contrast, the same DE pairs showed the opposite pattern of expression in the control samples (i.e. 60% of the same pairs had more 3'UTR transcripts), indicating that MC-LR changed the PCT/3'UTR ratio in about 20% of DE mRNAs (depending on the duration of exposure). This also indicated that, after MC-LR exposure, expression of our putative 3'UTR transcripts changed independently of PCT expression, which was also indicated by comparison of the pattern of expression of PCT/3'UTR pairs with the pattern of changes in expression of all PCT (paired and unpaired with 3'UTRs, Fig. 4).

Z komentarzem [AR65]: See my note in the Results section.

In the majority of cases, if a putative autonomous 3'UTR was DE after MC-LR exposure, the associated PCT was also DE in the same direction. Our results show that, out of all DE PCT/3'UTR pairs after 1d of MC-LR exposure, over 82% of the paired transcripts were upregulated (Fig. 3A). In contrast, at after 6 and 9d of exposure, there was no longer a majority of PCT/3'UTR pairs that were DE in the same direction. Moreover, gene ontology terms analysis showed that, after 1d of exposure, the co-upregulated pairs were enriched in transcription regulators, suggesting that these transcripts play roles in regulating the response to severe liver damage (Fig. 5). Additionally, our results show that DE PCT/3'UTR pairs at 6 and 9d of the exposure are similar in terms of function and direction of expression changes (Additional File 5). Interestingly, the differences in ratios of DE pairs between 1 and both 6 and 9d reflect differences observed in all DE PCTs (paired and unpaired) thus suggesting important participation of autonomous 3'-UTR in regulation of PCTs in response to MC-LR (Fig. 4). The observed changes in the liver transcriptome between the 1st and the 6th or 9th day of exposure may support our previous finding that challenging whitefish with MC-LR results in severe liver damage, which is

Z komentarzem [AR66]: See my notes in the Results for ideas about clarifying this.

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followed by resilience to further exposures to the toxin, allowing for regeneration of the damaged organ [14]. It should be investigated whether this apparent shift in the transcriptome profile reflects remodeling of liver cell processes for repair of the tissue.

Moreover, we also found that, in response to MC-LR exposure, some putative autonomous 3'UTRs and PCTs from the same mRNA were differently expressed in opposite directions, i.e. one was upregulated while the other was downregulated (Fig. 3A green and brown bars). We were particularly interested in pairs where the 3'UTR transcripts were upregulated and the PCTs were downregulated, as this could be attributed to a recently discovered mechanism in which the 3'UTR is cleaved and shortened [64]. The discoverers of that mechanism hypothesized that it serves as a global regulatory tool that works by increasing the effectiveness of miRNA binding sites upstream of the cleavage site. This would cause levels of 3'UTRs to rise after cleavage, while levels of the corresponding PCTs would drop as a result of more effective miRNA binding. Because our previous study showed that miRNAs that play roles in transcription regulation are aberrantly expressed after exposure to MC-LR [16], this hypothesis should be verified in the future. Furthermore, these results suggest the necessity of adopting a wider perspective when investigating the effects of MC-LR induced liver damage on NCTs and PCTs. Researchers looking to discover all aspects of transcriptome regulation by non-coding RNAs in MC-LR toxicity should also focus on decoding the crosstalk between NCTs and PCTs, as proposed by hypothesis of competing endogenous RNAs [65]. It has been shown recently that this strategy can be applied to successfully investigate mechanisms of MC-LR toxicity. For example, Meng et. al showed a transcriptomic regulatory network of miRNAs, piRNAs, circular RNAs, lncRNAs and mRNAs that were simultaneously involved in the cytotoxicity of MC-LR in

Z komentarzem [AR67]: The grammar here didn't make sense, so I had a quick look at the study by Malka et al. that you cite.

In their Discussion, those authors wrote "…we now show that thousands of mature poly(A) mRNAs, located mainly in the cytoplasm, are cleaved and polyadenylated post-transcriptionally at APA sites in their 3′-UTRs … Therefore, 3′-UTR shortening could be used as a global regulatory tool to increase the effectiveness of binding sites upstream of the cleavage point by shortening their distance from the transcript 3′ end."

Thus, it seems that Malka et al. demonstrated the mechanism and proposed what its function could be. So, in addition to fixing the grammar, I took the liberty of changing the meaning of your text in order to reflect what I understand from a quick reading of the relevant parts of Malka et al. Please double-check what I wrote and the paper by Malka et al.

For future reference, a hypothesis is "a supposition or proposed explanation made on the basis of limited evidence as a starting point for further investigation" (Oxford English Dictionary). Thus, hypotheses are proposed, suggested, etc., while findings are demonstrated, shown, etc.

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Z komentarzem [AR68]: The connection between your previous findings and the hypothesis of Malka et al. should be made clearer. Why does the fact that regulatory miRNAs are aberrantly expressed after MC-LR exposure provide a reason for verifying the hypothesis of 3′-UTR shortening serving as a global regulatory tool? In other words, your previous study is about miRNA expression, their hypothesis is about increasing the effectiveness of miRNA binding, and you need to connect those two ideas for the reader.

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Z komentarzem [AR69]: Make it clear whose results you are talking about, e.g. "the results of Malka et al. [64] suggest…"

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Z komentarzem [AR70]: This seems to be more than just a hypothesis. Kartha and Subramanian write that "Recently, several studies have demonstrated that both coding and non-coding RNA molecules can regulate gene expression in *trans* by acting as sponges of miRNAs (Karreth et al., 2011; ...

testicular tissues in mice [5].

Importantly, our discovery of the autonomous 3'UTRs in whitefish liver may be biased by the highly redundant nature of the *de novo* assembled genomes. Although we showed that the putative 3'UTR contigs were expressed independently of PCTs, some of the autonomous 3'UTRs paired with PCTs from the same mRNA could in fact be fragmented or incomplete mRNAs which were not assembled properly. However, even if a *de novo* transcriptome assembly approach for detecting various types of non-coding RNAs is not without flaws, it can be used as an extension to analyses based on annotated genomes. Because the detection of autonomous 3'-UTR transcripts using an annotated genome needs additional enrichment of 3'UTR transcripts in the RNA isolation or library preparation steps, it requires additional sequencing runs. Alternatively, pipelines designed to analyze autonomous 3'UTRs based on the *de novo* assembled transcriptome may serve as a backing strategy in preliminary research, eventually leading to more focused sequencing runs. Furthermore, already deposited transcriptomic data may be reused for additional *de novo* analysis by teams seeking direction for their studies on non-coding RNAs.

The aforementioned group of 84,974 putative novel lncRNA contained transcripts longer than 200 nucleotides with no homology to any tested database. We showed that MC-LR upregulated expression levels of 1739 putative novel lncRNAs and downregulated 2690 (Fig. 7). Comparing DE novel lncRNA transcripts common between days of exposure to MC-LR with the same groups of DE PCTs (Fig. 4) shows similar pattern in response to MC-LR. Besides already discussed potential role of lncRNAs in MC-LR-induced liver injury, this indicates differential co-expression of lncRNAs and mRNAs in MC-LR-induced liver injury in whitefish. Because as for

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Z komentarzem [AR71]: Starting the paragraph with "Importantly, one of our main findings may be based on biased data" seems to weaken the paper more than necessary. You can still discuss the limitations openly and honestly without emphasizing them so much. What about just starting with the second sentence? If anything important is lost by deleting the first sentence, you could just put it somewhere else in the paragraph.

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Z komentarzem [AR72]: This word doesn't seem to make sense here; unfortunately, I don't have an idea of what you're trying to say either.

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Z komentarzem [AR73]: Do you mean "the transcriptomic data that we have deposited"?

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Z komentarzem [AR74]: The grammar of sentence means that you are comparing transcripts to MC-LR. I'm guessing that you didn't want to say that you were comparing RNA to a microcystin, but I don't have any idea what you were trying to say.

Z komentarzem [AR75]: It's unclear what "this" refers to.

now it is the most common approach for identifying potential target genes of lncRNAs [12], we will further be able to identify potential target genes of lncRNAs and to investigate the biological function of lncRNAs in MC-LR-induced liver injury in whitefish. We further confirmed the change in expression of selected DE lncRNAs with qPCR, which validated our methodology (Additional File 6).

Z komentarzem [AR76]: What does “it” refer to?

Z komentarzem [AR77]: Unfortunately, I was unable to correct this part because problems with the English made it too difficult to understand.

These putative lncRNAs might serve as potential biomarkers for early detection of severe liver injury in whitefish (1d), as well as the fish’s recovery from exposure to the toxin, including regeneration of liver tissues (6d, 9d). As demonstrated in certain types of cancers [66], lncRNAs that have highly specific expression patterns and relatively stable secondary structures, and are efficiently detected in blood, plasma and urine have the potential to serve as novel noninvasive biomarkers for drug-induced liver injury. Our previous results suggest that the short, non-coding miRNA-122 can be a non-invasive biomarker for detecting liver damage in fish and a promising alternative to current gold-standard hepatotoxicity markers [67].

Z komentarzem [AR78]: necessary?

The adverse effects of MC-LR in whitefish are not limited only to the liver. For example, we previously showed that MC-LR caused brain injury in whitefish [68]. Although plasma levels of brain-specific MiR124-3p were not altered, it is possible that some brain-specific lncRNAs could serve as biomarkers of brain injury. Interestingly, MALAT1 shows relatively high expression in the brain, where it is involved in regulating synaptogenesis [69]. MALAT1 was downregulated in glioma [70] and is able to regulate levels of MiR124 in various diseases, including Parkinson’s disease [71]. Future studies will advance understanding of the roles of lncRNAs in MC-LR toxicity and will likely reveal novel biomarkers and targets for treatment. It will also be important to determine whether the aberrantly expressed lncRNAs detected in this study can be detected in noninvasively collected samples.

Conclusions

In this study, we detected differentially expressed polyadenylated lncRNAs in whitefish exposed to MC-LR. To achieve this, we constructed an extensive liver transcriptome that can be used to complete the current whitefish genome assemblies or to curate ones that are developed in the future. We obtained a dataset that provides a starting point for future studies on the role of lncRNAs in MC-LR–induced liver injury and subsequent liver regeneration. Among the detected DE transcripts, we identified novel, uncharacterized contigs that could potentially be used as non-invasive biomarkers of MC-LR–induced liver injury in whitefish. In addition, we detected transcripts with homology to lncRNAs previously described in other species. Current understanding of lncRNAs is limited [72], and we believe that our work is a step toward better understanding lncRNA expression and functions in the context of MC-LR toxicity, and the mechanisms of MC-LR toxicity in general. Lastly, we believe that to fully understand the molecular functions of lncRNAs, studies should adopt a broader perspective, including simultaneous analysis of all aspects of the network of competing endogenous RNAs. For that purpose, a combination of different methods of library preparation for RNA sequencing should be considered, which also may allow to uncover new RNA species.

Z komentarzem [AR79]: "…allow new types of RNA to be uncovered"?

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Acknowledgements

The project was funded by the National Science Centre of Poland (decision number: 2016/21/B/NZ9/03566).

Figures

Fig. 1 Schematic representation of pipeline used to profile changes in expression of putative long non-coding RNAs (lncRNAs) in MC-LR-induced liver damage in whitefish. Yellow shapes indicate pipeline input; green shapes indicate action step taken in analysis; blue shapes indicate output of an action; purple shapes indicate final output. Abbreviations: EXPLAIN ABBREVIATIONS IN THE FIGURE HERE

Fig. 2 Differentially expressed putative known non-coding RNAs after MC-LR exposure. (A-D) Volcano plots and Venn diagram of DE transcripts with homology to any transcript deposited in RFAM database (including lncRNAs). (E-H) Volcano plots and Venn diagram of DE transcripts with homology to transcripts labeled as lncRNAs in Rfam database. Expression of putative MALAT1 transcript was downregulated at 6d and 9d of MC-LR exposure.

Fig. 3 Pairs of DE putative autonomous 3' UTRs and PCTs from the same mRNA in whitefish liver after MC-LR exposure. (A) Percentages of DE contigs from the same mRNA that both were upregulated (red) or down-regulated (blue), as well as those with opposing expression profiles. (B) Percentages of contigs from the same mRNA for which the ratio of PCT to putative autonomous 3' UTR transcripts was higher in the control (PBS) than in the exposed (MC-LR)

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Z komentarzem [AR80]: This figure needs to have parts A and B explained separately.

Z komentarzem [AR81]: Figures should have abbreviations defined, even if you define them in the article itself. Not all journals require this, but it's always a good idea because many readers go straight to the figures without reading the article and reviewers don't like to search for an abbreviation if they forget what it means. DNA, mRNA and ANOVA do not need to be defined because they are so commonly used, and gene names also are not defined, but BUSCO, DE, MC-LR, lncRNA, PCT, etc. should all be defined. Remember to use the abbreviations in all figures after defining them. After Figure 1, you only need to define new abbreviations, then write "Other abbreviations as in Figure 1."

Note how Malka et al. do this in the Nature Communications article you cite (although they define TEX several lines after they first use it in the figure caption).

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groups (brown), as well as those for which the ratio was higher in the exposed than in the control groups (green).

Fig. 4 Percentage between days of exposure to MC-LR of putative novel, protein-coding and putative PCT/3'UTR pairs transcripts, that were either (A) upregulated or (B) downregulated. Similarities in percentages between all three types of transcripts indicate co-expression in response to MC-LR-induced liver injury.

Fig. 5 Gene ontology of co-expressed pairs of putative autonomous 3'UTRs and PCTs from the same mRNA that were upregulated after MC-LR exposure. Figure shows enriched gene ontology terms for pairs that were upregulated on d1 compared to d6 and d9.

Fig. 6 Distributions of length-corrected minimum free energy, content of GC base pairs and transcript lengths differ between PCTs and putative novel lncRNAs. (A) lncRNA transcripts have a higher mean length-corrected minimum free energy (−0.237 kcal/ mol/nt) than protein-coding transcripts (−0.289 kcal/ mol/nt); (B) lncRNA transcripts have a lower mean GC base pair content (0.393) than PCTs (0.460); (C) Comparison of distribution of lengths of transcripts. LncRNAs show shorter mean sequence length (949 bp) than that of protein-coding transcripts (1747 bp).

Fig. 7 MC-LR induced DE of putative novel lncRNAs in whitefish liver. Volcano plots after 1d (A), 6d (B) and 9d (C) of exposure. Venn diagrams of upregulated (D) and downregulated (E) putative novel lncRNAs after MC-LR exposure. Note that days 6 and 9 are more similar to each other than to day 1 in terms of which lncRNAs were DE on those days.

Z komentarzem [AR82]: As mentioned above, I don't understand this explanation or the figure, so I couldn't fully correct these sentences.

Z komentarzem [AR83]: Indicate the base of the logarithm on the horizontal axis, i.e. $\log_{10}$, $\log_2$, etc.

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Z komentarzem [AR84]: B and C are mixed up—you should change the figure or the caption accordingly.

Z komentarzem [AR85]: The mean is usually understood to indicate what a typical example of something looks like. Thus, it is not a good description of a typical transcript length when the distribution is strongly skewed, like that of the PCTs. Notice that the peaks of both distributions are very similar, meaning that the most common transcript lengths are similar. The difference is that there are many longer sequences in the PCT distribution. For this reason, it would be more accurate and statistically correct to write "Distribution of sequence lengths differs between lncRNAs and PCTs. Note that the distribution of PCT sequence lengths includes many longer sequence lengths."

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Fig. 8 Expression of putative MALAT1 (A), HOTAIR (B) and HOTTIP (C) in whitefish liver after 1d, 6d and 9d of MC-LR exposure (quantified with RT-qPCR).

Additional File 1 Histograms and normal Q-Q plots of minimum free energy (MFE), content of GC base pairs and sequence length in PCTs and NCTs.

Additional File 2 Detailed statistics on the quality of sequencing data from all (52) samples used for *de novo* assembly of the whitefish liver transcriptome. Samples in bold were used in this study for quantitative analysis of lncRNA.

Additional File 3 Contig metrics of the filtered *de novo* assembled liver transcriptome. All metrics were calculated using Transrate version 1.0.3.

Additional File 4 Alignment of our putative MALAT1 transcript to the seed sequence of human MALAT1 deposited in the Rfam database. (A) Output of cmscan hit alignment. (B) ClustalO alignment.

Additional File 5 Comparison of Gene Ontology terms of co-upregulated (A) and co-downregulated (B) putative PCT/3'UTR transcript pairs between 6d (red bars) and 9d (gray bars).

Additional File 6 Expression of selected putative novel lncRNAs in whitefish liver after 1d, 6d and 9d of MC-LR exposure quantified using Real-Time qPCR. Transcripts were selected based on RNA-Seq results.

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Z komentarzem [AR86]: This is the preferred term in the MIQE guidelines.

Z komentarzem [AR87]: Explain what the asterisks in the figure mean (I'm guessing it's p < 0.01), and define "Ctr".

Z komentarzem [AR88]: To save time (and your money), I'm checking the descriptions of the additional files, but not the files themselves unless I don't understand something.

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— **sformatowano:** Czcionka: Kursywa, Czcionka o złożonym skrypcie: Kursywa

Z komentarzem [AR89]: Should Gene Ontology be capitalized? Check for consistency throughout the manuscript—I wasn't sure what is preferred.

Z komentarzem [AR90]: As I understand it, you are comparing this data with the data from the control group. If this is correct, delete the word "between" and rewrite as "…pairs at 6d (red bars) and 9d (gray bars) of MC-LR exposure".

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